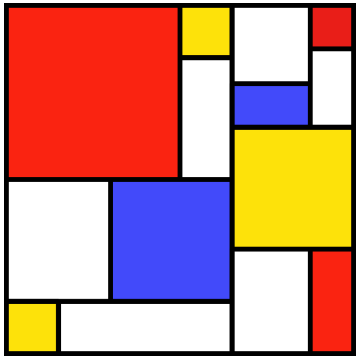
SLiM

Workshop Series



Nucleotide-based Models

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Nucleotide-based Models

- Tracks the nucleotides of every haplosome
- Mutations have an associated nucleotide
- Mutation rates are sequence-dependent
- Realistic gene conversion, including gBGC

```
GAATGTCGGTTAGAGCAACCTAGCTTCTCAGATCGCAATA
GAATGTCCGTTAGAGCAACCTAGCTTCTCAGATGGCTATA
GAATGTCCGTTAGAGCAACCTAGCTTCTCAGATGGCCATA
GAATGTCGGTTAGAGCATCCTAGCTTCTCAGATCGCAATA
GAATGTCGGTTAGAGCAACCTAGCTTCTCAGATCGCAATA
GAATGTCCGTTAGAGCAACCTAGCTTCTCAGATGGCAATA
GAATGTCGGTTAGAGCATCCTAGCCTCTCAGATGGCAATA
GAATGTCGGTTAGAGCATCCTAGCTTCTCAGATCGCAATA
```

Nucleotide-based Models

- Are **declared** as nucleotide-based
  - an option in `initializeSLiMOptions()`
- Have an **ancestral sequence** (reference sequence)
  - set with `initializeAncestralNucleotides()`
- Use nucleotide-based **mutation types**
  - set up with `initializeMutationTypeNuc()`
  - this establishes an `"l"` stacking policy
- Define sequence-based **mutation rate matrix**
  - passed to `initializeGenomicElementType()`

Ancestral sequence updating

- The ancestral sequence is the reference
  - used whenever a mutation is not present
- It updates whenever a mutation fixes

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```
GAATG

GAATG
GAATG
GAATG
GAATG
```

Ancestral sequence updating

- The ancestral sequence is the reference
  - used whenever a mutation is not present
- It updates whenever a mutation fixes

```
GAATG      GAATG

GAATG      GACTG
GAATG  ➡   GAATG
GAATG      GAATG
GAATG      GAATG
```

Ancestral sequence updating

- The ancestral sequence is the reference
  - used whenever a mutation is not present
- It updates whenever a mutation fixes

```
GAATG      GAATG      GAATG

GAATG      GACTG      GACTG
GAATG  ➡   GAATG  ➡   GACTG
GAATG      GAATG      GACTG
GAATG      GAATG      GAATG
```

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```
GAATG        GAATG        GAATG        GAATG

GAATG        GACTG        GACTG        GACTG
GAATG   ➡    GAATG   ➡    GACTG   ➡    GACTG
GAATG        GAATG        GACTG        GACTG
GAATG        GAATG        GAATG        GACTG
```

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GAATG      GAATG      GAATG      GAATG      GACTG

GAATG      GACTG      GACTG      GACTG      GACTG
GAATG  ➡   GAATG  ➡   GACTG  ➡   GACTG  ➡   GACTG
GAATG      GAATG      GACTG      GACTG      GACTG
GAATG      GAATG      GAATG      GACTG      GACTG
```

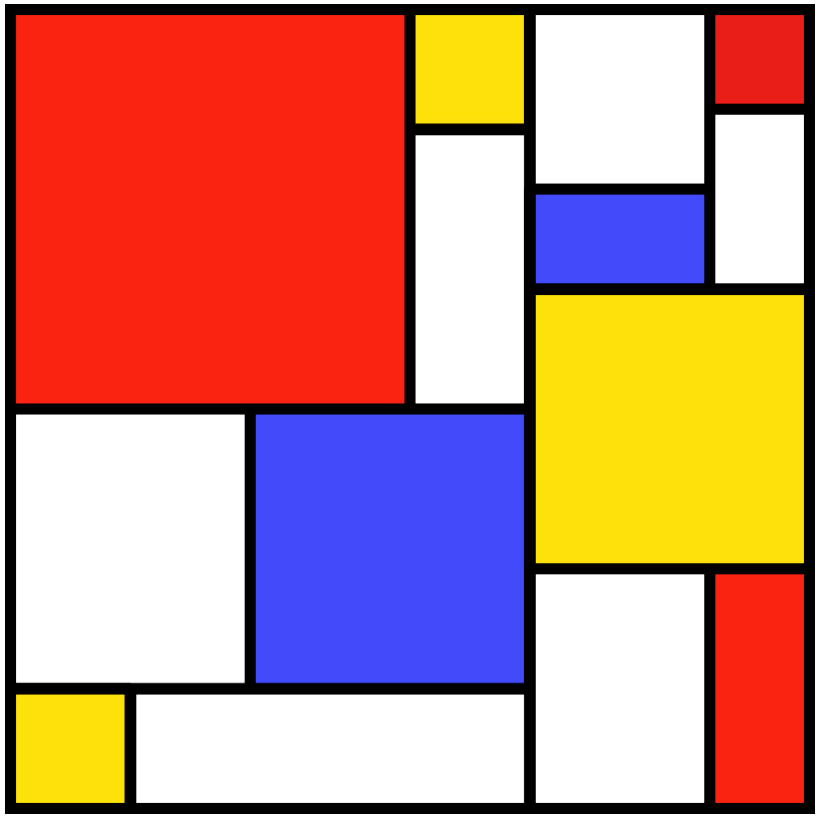
Nucleotide-based Models

- Optional facilities
  - Trinucleotide-based mutation rates
  - Reading and writing FASTA, VCF files
  - Getting the nucleotide sequence
  - Getting the codon sequence
  - Getting the amino acid sequence
  - Hotspot maps (variable mutation rate)
  - GC-biased gene conversion (gBGC)

A complete nucleotide-based model

```
initialize() {
    defineConstant("L", 1e6);
    initializeSLiMOptions(nucleotideBased=T);
    initializeAncestralNucleotides(randomNucleotides(L));
    initializeMutationTypeNuc("m1", 0.5, "f", 0.0);
    initializeGenomicElementType("g1", m1, 1.0, mmJukesCantor(1e-7));
    initializeGenomicElement(g1, 0, L-1);
    initializeRecombinationRate(1e-8);
}
1 early() { sim.addSubpop("p1", 500); }
2000 late() { sim.outputFixedMutations(); }
```

- Declared as nucleotide-based
- Sets the ancestral (reference) sequence
- Uses a nucleotide-based `MutationType`
- Defines a mutation rate matrix



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